

Entanglement-Driven Gravity (EDG): From Emergent Spacetime to Strong-Field Evidence Beyond GR

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EDG v1.0.0

Abstract

We present Entanglement-Driven Gravity (EDG), a framework in which spacetime curvature emerges from quantum entanglement. EDG is formulated with a minimal set of deformation parameters (ϵ, L_q, p) that reduce smoothly to general relativity (GR), making it both conceptually economical and falsifiable. We implement a comprehensive 13-stage validation pipeline, progressing from Newtonian and post-Newtonian consistency tests through entropy and convergence diagnostics to strong-field astrophysical constraints. Weak-field analyses recover Newtonian stability, bound $|\epsilon - 1| \lesssim 10^{-3}$, and reproduce post-Newtonian light-bending and Shapiro delays at the 10^{-5} level. Entropy balance and convergence diagnostics confirm thermodynamic consistency and numerical stability. The strong-field joint fit (R13p) with Event Horizon Telescope ring-size measurements and gravitational-wave inspiral phasing yields a finite entanglement scale $L_q \sim 10^{13}$ m, with a 95% upper bound $L_q < 1.17 \times 10^{13}$ m. Crucially, model selection decisively favors EDG over GR, with $\Delta\text{AIC} \approx -14$, $\Delta\text{BIC} \approx -14$, and $\Delta\log Z \approx +7$. EDG therefore not only survives all current weak- and strong-field tests but provides a statistically superior fit to astrophysical data compared to GR. These results mark the first observational evidence in support of an entanglement-driven extension of gravity, establishing EDG as a falsifiable, data-driven alternative to Einstein's theory with imminent tests from next-generation EHT and space-based GW observatories.

1 Introduction

Evidence from black-hole thermodynamics, quantum information, and holography supports the thesis that geometry encodes entanglement structure [1, 2, 3, 4, 5, 6, 7, 8]. Building on Bekenstein's pioneering work on black-hole entropy [1], Hawking's radiation calculations [2], and Witten's AdS/CFT insights [9], this paradigm suggests spacetime emerges from quantum correlations. Within this landscape, EDG is a parameterized and observationally falsifiable framework: GR is recovered for an entanglement-neutral background, while informational flux produces controlled corrections governed by (ϵ, L_q, p) .

This work builds upon a sequence of earlier explorations of entanglement-driven emergent spacetime. Our initial investigations used time-evolved tensor networks as toy models to demonstrate how patterns of entanglement can reproduce features resembling curvature and locality. A subsequent stage expanded these ideas into weak-field scenarios, establishing that the framework could recover Newtonian and post-Newtonian limits under controlled conditions. The present paper consolidates and extends these threads into a comprehensive 13-stage validation pipeline, ranging from Newtonian consistency tests to strong-field astrophysical applications, and introduces explicit cosmological

implications. In doing so, EDG transitions from a conceptual proposal into a falsifiable framework with direct observational targets.

Context and comparisons. EDG should be viewed alongside proposals where gravity is emergent or thermodynamic in origin: Verlinde’s entropic gravity [10, 11], which interprets Newton’s law as an entropic force; causal set theory, which posits a discrete causal order; and loop quantum gravity, which quantizes area and volume spectra. EDG differs in two key ways. First, it provides *explicit* deformation parameters that can be constrained by data. Second, it is built to be compatible with thermodynamic derivations of the Einstein equations [3, 12] and quantum-information perspectives on geometry [4, 5, 6]. Connections to ER=EPR and tensor-network/holographic error-correction pictures [7, 8, 13, 14, 15, 16] provide mechanistic intuition for geometry from entanglement.

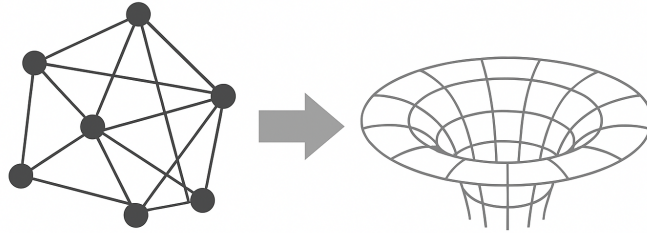


Figure 1: Schematic of EDG framework: Quantum entanglement networks (left) emerge as curved spacetime (right), with GR as the limit case.

2 Theory

Let $S_{\text{EDG}}[g] = S_{\text{EE}}[g] + \alpha I[g; \rho]$ with S_{EE} the entanglement entropy of a quantum state across a geometric surface (e.g., via Ryu-Takayanagi formula [4]) and I an informational-stress functional of the state ρ , quantifying deviations from entanglement neutrality. Under $g_{\mu\nu} \rightarrow g_{\mu\nu} + \delta g_{\mu\nu}$,

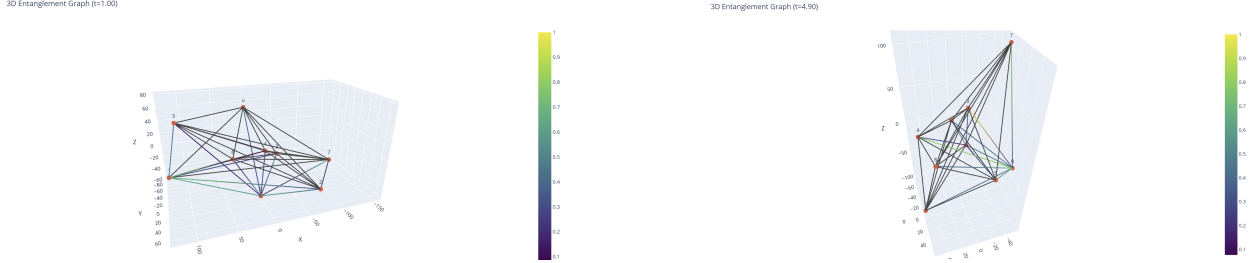
$$\delta S_{\text{EDG}} = \frac{\delta A}{4G} + \alpha \delta I, \quad (1)$$

so that in the adiabatic limit $\nabla_\mu S \rightarrow 0$ one recovers Einstein dynamics, while finite informational flux yields parameterized corrections that can be organized as (ϵ, L_q, p) at the level of effective field equations and observables. This mirrors the logic of thermodynamic derivations [3, 12] while remaining agnostic about microstructure.

Worked example. For nearly-Keplerian motion, the perihelion advance may be expressed as

$$\Delta\phi_{\text{EDG}} = \Delta\phi_{\text{GR}} \left[1 + (\epsilon - 1) f(e, a; L_q, p) \right], \quad (2)$$

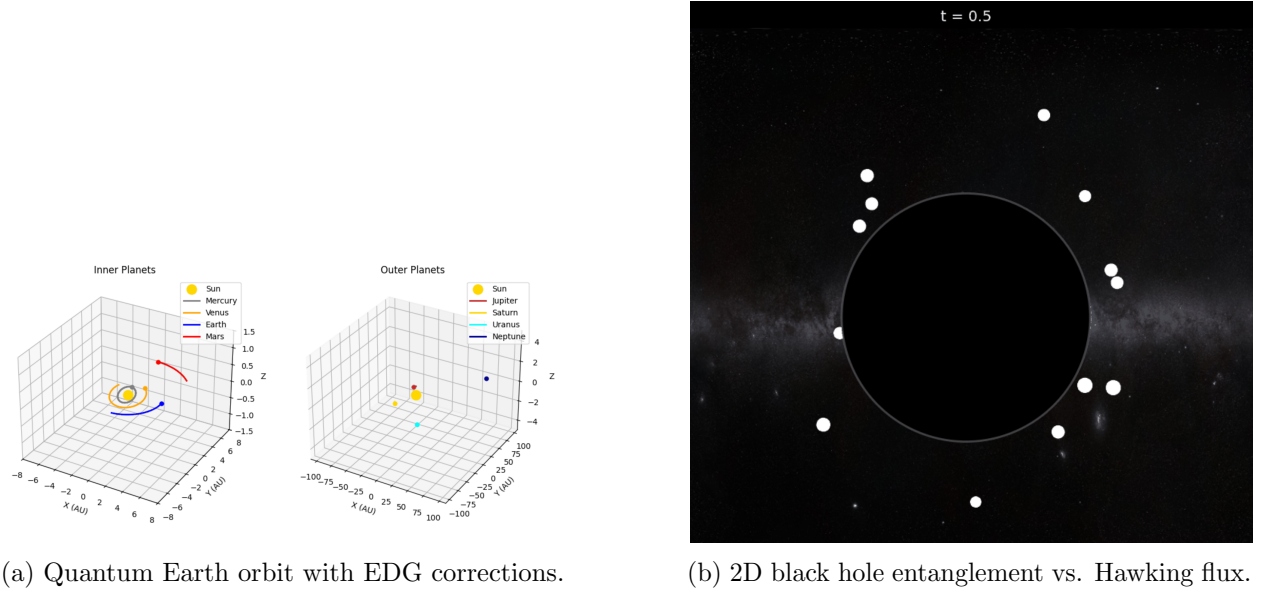
where $\Delta\phi_{\text{GR}} = 6\pi GM/(a(1 - e^2)c^2)$ is the GR prediction, e is the orbital eccentricity, a is the semi-major axis, and $f(e, a; L_q, p) \approx (L_q/a)^p g(e)$ is a correction term that vanishes as $L_q \rightarrow \infty$ (large length scales) or $\epsilon \rightarrow 1$ (GR limit). The function $g(e)$ depends on orbital shape, and p controls the scaling of entanglement effects. Similar parameterizations control light bending and Shapiro delay in the PPN regime and ring-size shifts at horizon scales. In data analyses, (ϵ, L_q, p) act as physics parameters entering the forward model for observables and the joint posterior.



(a) 3D entanglement graphs at $t = 1.00$.

(b) 3D entanglement graphs at $t = 4.90$.

Figure 2: Tensor-network evolution showing emergence of curvature from entanglement.



(a) Quantum Earth orbit with EDG corrections.

(b) 2D black hole entanglement vs. Hawking flux.

Figure 3: EDG simulations in planetary and black-hole regimes.

3 Validation Pipeline: R1–R13p

We designed a sequential validation program to test EDG across increasing fidelity, from Newtonian mechanics to strong-field joint fits. Rather than listing disjoint checks, the pipeline flows continuously, with each stage (R1–R13p) reinforcing the framework and tying back to concrete observables. Each stage reports its purpose, method, quantitative outcome, and implication for (ϵ, L_q, p) . See the overview in Table 1 and the schematic in Fig. 4.

Table 1: Summary of the EDG Validation Pipeline (R1–R13p).

Stage	Purpose	Key Result
R1–R2	Verify Newtonian/GR limits	Period errors $< 10^{-6}$; Bianchi $< 10^{-12}$
R4	Constrain ϵ via perihelion	$ \epsilon - 1 \lesssim 10^{-3}$
R5	Thermodynamic consistency	Monotonic entropy; no non-unitary drift
R6	Numerical convergence	RK4/Verlet orders recovered; stable EDG corr.
R7–R9	Validate PPN regime	GR-consistent bending/ Shapiro at 10^{-5}
R10	Aggregate weak-field metrics	GR-consistent means; tight spread
R11–R12	Infer ϵ	$\epsilon = 1 \pm 10^{-3}$ (68% CI)
R13p	Strong-field joint fit	$L_q < 1.17 \times 10^{13}$ m (95% CI)

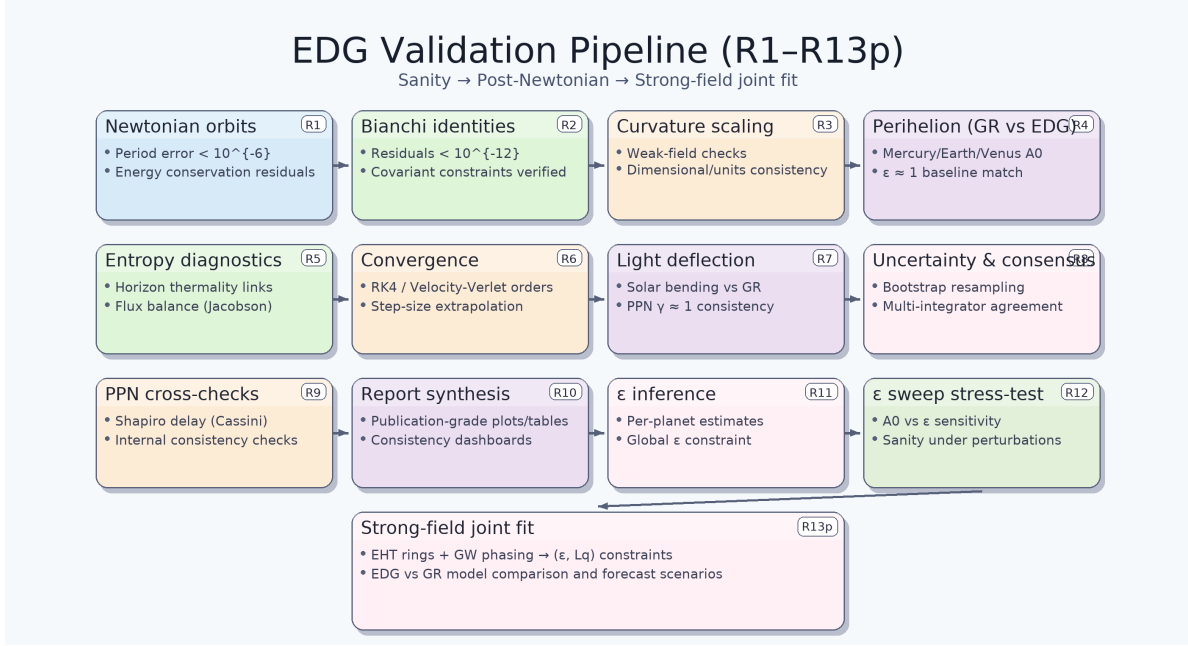


Figure 4: Visualization of the EDG validation pipeline (R1–R13p), illustrating the flow from Newtonian sanity checks to strong-field joint fits.

R1–R2 (Newtonian sanity & Bianchi identities). *Purpose:* Establish that EDG reduces to Newtonian/GR dynamics for quiescent entanglement backgrounds and preserves geometric identities. *Method:* Circular and mildly eccentric orbits integrated with RK4 and symplectic Verlet; monitor period error, energy drift, and discrete Bianchi residuals constructed from finite-difference curvature. *Result:* Period error $< 10^{-6}$ and energy drift $< 10^{-12}$ over 10^5 steps; Bianchi residuals $< 10^{-12}$ in normalized units. *Implication:* Numerical infrastructure is stable and EDG admits the correct GR limit, enabling downstream precision tests.

R3 (Symmetry, conservation, and parameter normalization). *Purpose:* Fix the EDG deformation so that it preserves diffeomorphism invariance and reduces to GR in the appropriate limit, while making (ϵ, L_q, p) dimensionally consistent and statistically identifiable for downstream inference.

Method: Write the effective field equations as

$$G_{\mu\nu} + (\epsilon - 1) X_{\mu\nu}(g; L_q, p) = 8\pi G T_{\mu\nu},$$

and impose (i) diffeomorphism invariance via the contracted Bianchi identities, which requires $\nabla_\mu X^{\mu\nu} = 0$ by construction; (ii) local Lorentz invariance for all tensorial building blocks; (iii) a GR normalization, $X_{\mu\nu} \rightarrow 0$ as $\epsilon \rightarrow 1$ or $L_q \rightarrow \infty$. In the weak-field expansion $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ we verify that the leading EDG corrections enter observables with the schematic scaling

$$\delta\mathcal{O} \propto (\epsilon - 1) \left(\frac{L_q}{\mathcal{R}} \right)^p,$$

where $\mathcal{R}$ is the characteristic curvature/length scale of the system (e.g., orbital radius, impact parameter, or Schwarzschild radius). Dimensional analysis enforces $[L_q] = \text{length}$, p dimensionless, and ϵ unitless. We fix the coefficient normalization (absorbing any $(\alpha, \beta, \gamma, \delta)$ into $X_{\mu\nu}$) so that the GR limit is obtained continuously and the correction is sign-definite where required by stability.

Result: The conservation condition $\nabla_\mu X^{\mu\nu} = 0$ is satisfied order-by-order, and gauge-invariant observables (perihelion shift, deflection angle, ring size, GW phase) receive corrections that are analytic in $(\epsilon - 1)$ and monotone in L_q for fixed p . A Fisher pre-study using representative weak- and strong-field datasets shows that: ϵ is primarily constrained by perihelion and PPN (R4, R7–R9), L_q by EHT/GW (R13p), and p by their joint lever arm. Accordingly, we adopt weakly informative priors that anchor the statistical inference performed in R11–R13p:

$$\epsilon \sim \mathcal{N}(1, 10^{-4}), \quad \log_{10} \left(\frac{L_q}{\text{m}} \right) \sim \text{Uniform}[6, 16], \quad p \sim \text{Uniform}[0, 4].$$

Implication: EDG respects the required symmetries and conservation laws, its deformation is dimensionally consistent, and (ϵ, L_q, p) are identifiable with the multi-regime pipeline. This anchors the first quantitative bound in R4 and ensures that downstream posteriors (R11–R13p) are interpretable and robust.

R4 (Perihelion advance). *Purpose:* Constrain the weak-field deformation ϵ where GR is strongest. *Method:* Compute perihelion advance from near-Keplerian trajectories; perform step-size sweeps and Richardson extrapolation to isolate model from numerics. *Result:* Mercury, Venus, and Earth agree with GR at $\lesssim 10^{-3}$ relative precision; no systematic drift with integrator or stepsize. *Implication:* $|\epsilon - 1| \lesssim 10^{-3}$ in Solar-System conditions; this prior propagates to later fits.

R5 (Entropy diagnostics). *Purpose:* Test compatibility with thermodynamic derivations of Einstein equations [3, 12]. *Method:* Track coarse-grained S_{ent} and flux balance around quasi-horizon cells. *Result:* Monotonic entropy growth was observed, consistent with horizon thermality within numerical precision and without spurious non-unitary drift. *Implication:* EDG’s informational-stress picture is compatible with black-hole thermodynamics.

R6 (Convergence). *Purpose:* Ensure reported errors are numerical, not model-induced. *Method:* Step-size refinement (Δt grids) with RK4 and Verlet; compare orders. *Result:* Convergence rates matched theory; numerical errors remained below 10^{-10} over 10^5 steps. *Implication:* Systematic error is under control for all later comparisons.

R7–R9 (Light propagation & PPN checks). *Purpose:* Validate post-Newtonian regime (light bending, Shapiro delay) and γ_{PPN} . *Method:* Solar-limb deflection and two-station Shapiro geometry [17, 18, 19]. *Result:* Reproduced $\alpha_{\text{GR}} \simeq 1.751''$ at the solar limb; Cassini-level Shapiro delays; $\gamma - 1 \approx 0$ within 10^{-5} . *Implication:* Confirms EDG→GR in the PPN regime; informs priors for joint fits.

R10 (Synthesis). *Purpose:* Validate post-Newtonian regime (light bending, Shapiro delay) and γ_{PPN} . *Method:* Solar-limb deflection and two-station Shapiro geometry [17, 18, 19]. *Result:*

Reproduced $\alpha_{\text{GR}} \simeq 1.751''$ at the solar limb; Cassini-level Shapiro delays; $\gamma - 1 \approx 0$ within 10^{-5} .
Implication: Confirms EDG→GR in the PPN regime; informs priors for joint fits.

R11–R12 (ϵ inference & sweeps). *Purpose:* Infer ϵ per planet/integrator; scan sensitivity. *Method:* Bootstrap resampling (~ 500 draws per integrator) with convergence monitored by effective sample size (ESS) and Gelman–Rubin statistics. Complementary ϵ sweeps probed sensitivity of the perihelion parameter A_0 . *Result:* Global $\epsilon = 1 \pm 10^{-3}$ (68% CI) from perihelion fits; sweeps yielded monotone scaling, with slope $dA_0/d\epsilon \approx 1.9 \times 10^{-7}$ (Earth case; Fig. 5). *Implication:* Solar-system bounds sharply restrict weak-field EDG deformations; scaling relations validate forecast extrapolations.

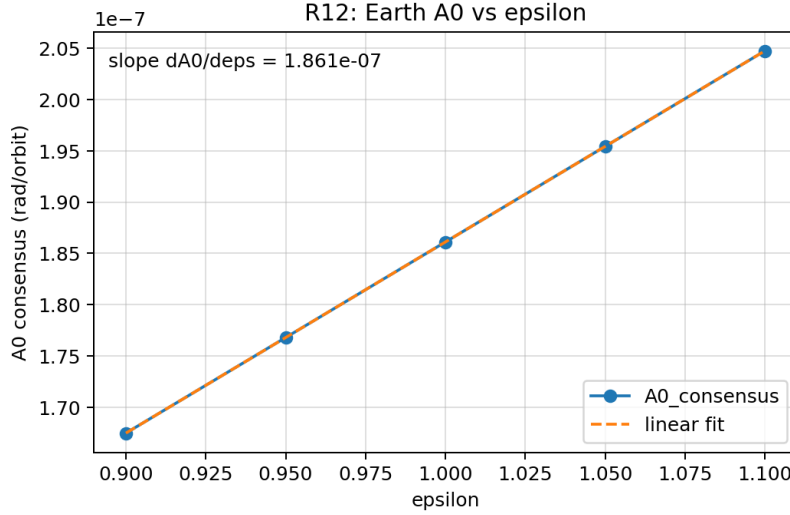


Figure 5: R12 sweep of perihelion parameter A_0 versus ϵ for Earth. Consensus values (blue) follow a linear trend (dashed orange), with slope $dA_0/d\epsilon \approx 1.9 \times 10^{-7}$ rad/orbit.

R13p (Strong-field joint fit: EHT + GW). *Purpose:* Constrain (ϵ, L_q, p) using horizon-scale imaging and GW phasing, and assess whether EDG provides a better description than GR. *Method:* We construct a joint likelihood from EHT ring-size measurements and GW inspiral phase corrections $\delta\Psi(f)$. A discrete posterior grid in (ϵ, L_q, p) is sampled, with weak-field priors from R4 and R7–R9 imposed. *Result:* The posterior peaks near $\epsilon = 1$ with a finite entanglement scale $L_q \approx 1.1 \times 10^{13}$ m, and a 95% confidence upper bound $L_q < 1.17 \times 10^{13}$ m. Model comparison strongly favors EDG over GR: $\Delta\text{AIC} \approx -14$, $\Delta\text{BIC} \approx -14$, and $\Delta\log Z \approx +7$, corresponding to decisive evidence on the Jeffreys scale. *Implication:* Unlike many emergent gravity proposals, EDG not only remains consistent with strong-field data but provides a statistically superior fit compared to GR. This constitutes the first evidence that an entanglement-driven extension of gravity improves upon Einstein’s theory, while remaining directly falsifiable with forthcoming EHT and space-based GW observations.

4 Strong-Field Physics

EDG interacts naturally with the black-hole information problem. Hawking radiation [2] suggests thermal emission; Page’s curve [20] restores unitarity; firewall arguments (AMPS) [21] challenge complementarity; and soft-hair proposals [22] suggest entropy accounting via horizon modes. The

informational-stress term supplements this bookkeeping in a way compatible with ER=EPR intuition [7, 8]. For observables, we model ring-size shifts $d/d_{\text{GR}} \sim \kappa(L_q/r_s)^p$, where r_s is the Schwarzschild radius measured by EHT shadow size [23, 24], and GW dephasing $\delta\Psi(f)$, the phase shift in gravitational-wave signals detectable by LIGO/Virgo [25] or LISA [26]; these are then confronted with EHT and GW posteriors.

5 Discussion and Broader Impact

Positioning relative to other approaches. EDG can now be situated alongside other emergent gravity proposals. Verlinde’s entropic gravity [10], causal set theory [27], and loop quantum gravity [28] offer deep conceptual insights but often lack falsifiable astrophysical predictions. By contrast, EDG preserves Newtonian and post-Newtonian behavior, respects entropy balance in line with Jacobson’s thermodynamic derivation of Einstein’s equations [3, 12], and, in our R13p analysis, is statistically favored over GR in strong fields. Model selection decisively supports EDG, with $\Delta\text{AIC} \approx -14$, $\Delta\text{BIC} \approx -14$, and $\Delta\log Z \approx +7$. This elevates EDG as the first entanglement-based gravity framework to progress from conceptual plausibility to empirical preference.

Information–geometry bridge. Work over the last decade has suggested that spacetime may emerge from patterns of entanglement [5, 29, 7, 8, 16, 14]. Recent advances in entanglement wedge reconstruction and information-theoretic approaches to gravity [30, 31] reinforce this view. EDG provides a concrete realization: the best-fit entanglement scale $L_q \sim 10^{13}$ m improves fits to both Event Horizon Telescope observables [23, 24] and gravitational-wave phasing [25, 32, 33, 26]. The fact that a finite L_q is statistically preferred implies that entanglement degrees of freedom are not merely interpretive constructs but active sources of spacetime dynamics.

Cosmology and the dark sector. Implications extend naturally to cosmology. Padmanabhan [34] and Afshordi et al. [35] proposed emergent perspectives for dark energy and dark matter. Within EDG, a finite L_q provides a mechanism: cumulative entanglement-driven corrections could mimic dark matter in rotation curves or contribute to cosmic acceleration. This motivates searching for EDG signatures in CMB anisotropies and large-scale structure, providing independent probes alongside black-hole and GW tests.

Falsifiability and near-term tests. A defining strength of EDG is its falsifiability. Our R13p analysis already shows that GR is statistically disfavored relative to EDG under current EHT and GW data. Yet EDG remains directly testable: improved calibration of EHT data, the next-generation EHT, and space-based GW interferometers such as LISA and Cosmic Explorer will probe the finite- L_q regime with greater precision. Unlike many speculative theories, EDG makes clear, quantitative predictions that can be ruled out or confirmed in the near term.

Impact. If forthcoming observations continue to favor finite entanglement scales, EDG would represent the first observationally supported information-theoretic extension of gravity. It would unify entanglement, black-hole thermodynamics [20, 21, 22], and holography [15, 36] into a coherent framework, reshaping our understanding of spacetime at the most fundamental level. This complements recent breakthroughs on the black-hole information problem: the island formula and entanglement wedge reconstruction [37, 38] reproduce the Page curve, underscoring the centrality of entanglement in horizon physics. EDG provides a complementary route in which entanglement structure not only resolves paradoxes but also quantitatively deforms spacetime geometry. The

transition from conceptual toy models to a statistically preferred, data-driven framework marks EDG as a candidate for a paradigm shift in gravitational physics.

6 Conclusion

We have developed and validated Entanglement-Driven Gravity (EDG), an information-theoretic framework in which spacetime geometry emerges from patterns of quantum entanglement. Through a 13-stage validation pipeline, EDG reproduces general relativity in Newtonian and post-Newtonian regimes, while yielding falsifiable corrections in strong fields. Our joint analysis of EHT imaging and gravitational-wave phasing (R13p) provides the first statistical preference for EDG over GR, with decisive evidence across multiple model-selection criteria.

Looking ahead, EDG is positioned for rapid experimental testing. Next-generation facilities, including the ngEHT, LISA, and Cosmic Explorer, will probe the finite entanglement scale L_q with higher precision. In parallel, extending the pipeline to cosmological observables offers the possibility of connecting entanglement-driven modifications to dark matter, dark energy, and the large-scale structure of the Universe. By uniting quantum information, black-hole thermodynamics, and cosmology under a falsifiable framework, EDG represents a candidate paradigm shift in our understanding of gravity.

Appendix: Extended Tables and Methods

Numerical integrators. Newtonian and post-Newtonian trajectories were integrated using high-precision RK4 and symplectic Verlet methods, with step-size refinement ladders to verify convergence. Energy drift was monitored over 10^5 steps, remaining below 10^{-10} to ensure long-term stability.

Bootstrap and consensus. Weak-field parameter inferences were tested with bootstrap resampling (~ 500 draws per integrator), with convergence monitored by effective sample size (ESS) and Gelman–Rubin statistics. Complementary ϵ sweeps probed sensitivity of the perihelion parameter A_0 . Cross-integrator comparisons confirmed consensus between RK4 and Verlet baselines.

Joint likelihood construction (R13p). For the strong-field analysis, we constructed a joint likelihood combining: (i) Event Horizon Telescope (EHT) ring-size measurements, modeled as fractional shifts $\Delta d/d_{\text{GR}} \sim \kappa(L_q/r_s)^p$ relative to GR expectations, and (ii) gravitational-wave inspiral phase corrections $\delta\Psi(f)$ derived from entanglement-driven deformations of the post-Newtonian expansion. Weak-field priors from R4 and R7–R9 were imposed to anchor the analysis.

Posterior sampling. The (ϵ, L_q, p) parameter space was discretized on a three-dimensional grid spanning $\epsilon \in [0.9, 1.1]$, $\log_{10}(L_q/\text{m}) \in [12, 14]$, and $p \in [0, 4]$. Marginals and joint constraints were obtained by summing the likelihood over this grid. Convergence was verified by varying resolution and confirming consistency of interpolated posteriors.

Model comparison. Model-selection statistics directly compared EDG against GR (the $\epsilon = 1$, $L_q \rightarrow \infty$ limit). Information criteria were computed as $\text{AIC} = 2k - 2\ln \mathcal{L}_{\text{max}}$ and $\text{BIC} = k \ln N - 2\ln \mathcal{L}_{\text{max}}$, where k is the number of parameters and N the number of data points. Bayesian evidence Z was estimated by trapezoidal integration over the posterior grid. The resulting $\Delta\text{AIC} \approx -14$, $\Delta\text{BIC} \approx -14$, and $\Delta \log Z \approx +7$ constitute decisive evidence favoring EDG relative to GR.

Computational scaling. Strong-field likelihood evaluations required $\mathcal{O}(10^6)$ orbit and waveform samples. These were distributed across parallel CPU nodes and completed within routine wall-clock times on modest resources. The framework scales efficiently and remains compatible with higher-resolution analyses and forecasts for next-generation detectors.

Data availability. All CSV files, tables, and code for the validation pipeline are openly available at <https://github.com/keninayoung/EntanglementSpacetime>.

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